

GENERAL SPECIFICATION G1.1

Safety in Design

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## **G1.1 SAFETY IN DESIGN**

### **G1.1.1 Design for Safety**

- (a) The Contractor shall ensure that any design completed is performed in accordance to the Guidelines on Design for Safety in Buildings and Structures (the "Guidelines") published by the WSH Council. The Contractor shall refer to WSH's website for the latest guidelines.

### **G1.1.2 Safety Features**

- (a) The Works as installed shall have no features that could be a hazard to operators, maintenance staff or other persons having access thereto. Guards, electrical safety devices, thermal insulation, noise suppression devices, written notices, safety signs, and safety colours shall be provided where needed.
- (b) Adequate safeguards shall be provided to prevent personnel accidentally coming in contact with dangerous machinery, mechanisms, hot surfaces, electrically live parts and other hazardous components or contents.
- (c) Emergency stop devices shall be provided and positioned adjacent to all driven plant which:
  - (i) Involves a risk of injury to personnel during normal or maintenance operations;
  - (ii) Handles high pressure fluid or dangerous chemicals;
  - (iii) Has no direct or unimpeded access between the controller and the plant; and
  - (iv) Is not visible to its controller or is located at a distance of more than ten metres from it.
- (d) Nothing in this Specification shall remove the Contractor's obligation to provide a safe installation.

### **G1.1.3 Hazard and Operability (HAZOP) Studies**

- (a) The Hazard and Operability (HAZOP) Studies shall be performed in accordance with the recommendations in the latest edition of IEC 61882 Hazard and operability studies (HAZOP studies) – Application Guide or any equivalent standard. The study shall be applied to the process and instrumentation diagrams produced under the Contract prior to the submission of the same for approval.
- (b) The HAZOP study sessions shall be held at the S.O.'s site office and shall be attended by the Contractor's project manager and the persons responsible for the design of the subject matter. The Contractor shall appoint one of his team to act as secretary responsible for recording, typing and issuing copies of the minutes to each person present at the meeting. A representative of the S.O. will act as chairman.

**END OF SECTION**